

Vector C Intrinsics Reference Document

Valid Values of SEW and LMUL:

```
1 SEW = 8, 16, 32
2 LMUL = m1, m2, m4, m8, mf2, mf4, mf8 // 'f' denotes fractional LMUL
```

Vector Datatype Prototypes:

```
1 vint{SEW}{LMUL}_t //Vector of signed integers
2 vuint{SEW}{LMUL}_t //vector of unsigned integers
```

Configuration Instructions:

```
1 //Configure vectors of length avl with selected configuration
2 size_t __riscv_vsetvl_e{SEW}{LMUL}(size_t avl)
3 //Configure vectors of length VLMAX with selected configuration
4 size_t __riscv_vsetvlmax_e{SEW}{LMUL}()
```

Load-Store Prototypes:

```
1 //Unit-Stride Load/Store
2 vint{SEW}{LMUL}_t __riscv_vle{SEW}_v_i{SEW}{LMUL}(const int{SEW}_t *rs1, size_t vl);
3 void __riscv_vse{SEW}_v_i{SEW}{LMUL}(int{SEW}_t *rs1, vint{SEW}{LMUL}_t vs, size_t vl);
4 //Strided Load/Store
5 vint{SEW}{LMUL}_t __riscv_vlse{SEW}_v_i{SEW}{LMUL}(const int{SEW}_t *rs1, ptrdiff_t rs2, size_t vl);
6 void __riscv_vsse{SEW}_v_i{SEW}{LMUL}(int{SEW}_t *rs1, ptrdiff_t rs2, vint{SEW}{LMUL}_t vs, size_t vl);
```

Function Prototypes:

```
1 //Single Width Arithmetic Functions
2 vint{SEW}{LMUL}_t __riscv_{FUNC}_vv_i{SEW}{LMUL}(vint{SEW}{LMUL}_t vs2, vint{SEW}{LMUL}_t vs1,
3 size_t vl);
4 vint{SEW}{LMUL}_t __riscv_{FUNC}_vx_i{SEW}{LMUL}(vint{SEW}{LMUL}_t vs2, int{SEW}_t rs1, size_t vl);
5 //Widening Arithmetic Functions
6 vint{SEW*2}{LMUL*2}_t __riscv_{FUNC}_vv_i{SEW*2}{LMUL*2}(vint{SEW}{LMUL}_t vs2, vint{SEW}{LMUL}_t vs1,
7 size_t vl);
8 vint{SEW*2}{LMUL*2}_t __riscv_{FUNC}_vx_i{SEW*2}{LMUL*2}(vint{SEW}{LMUL}_t vs2, int{SEW}_t rs1,
9 size_t vl);
```

Valid Functions:

```
1 vadd, vsub, vmul //Single Width
2 vwadd, vwsub, wvmul //Widening
```

Special Function Prototypes:

```
1
2 //Sign Extend
3 vint{SEW*2}{LMUL*2}_t __riscv_vsext_vf2_i{SEW*2}{LMUL*2}(vint{SEW}{LMUL}_t vs2, size_t vl);
4 vint{SEW*4}{LMUL*4}_t __riscv_vsext_vf4_i{SEW*4}{LMUL*4}(vint{SEW}{LMUL}_t vs2, size_t vl);
5 //Single Width Multiply-Accumulate
6 vint{SEW}{LMUL}_t __riscv_vmac_vv_i{SEW}{LMUL}(vint{SEW}{LMUL}_t vd, vint{SEW}{LMUL}_t vs1,
7 vint{SEW}{LMUL}_t vs2, size_t vl);
8 vint{SEW}{LMUL}_t __riscv_vmac_vx_i{SEW}{LMUL}(vint{SEW}{LMUL}_t vd, int{SEW}_t rs1,
9 vint{SEW}{LMUL}_t vs2, size_t vl);
10 //Widening Multiply-Accumulate
11 vint{SEW*2}{LMUL*2}_t __riscv_vvmac_vv_i{SEW*2}{LMUL*2}(vint{SEW*2}{LMUL*2}_t vd,
12 vint{SEW}{LMUL}_t vs1, vint{SEW}{LMUL}_t vs2, size_t vl);
13 vint{SEW*2}{LMUL*2}_t __riscv_vvmac_vx_i{SEW*2}{LMUL*2}(vint{SEW*2}{LMUL*2}_t vd, int{SEW}_t rs1,
14 vint{SEW}{LMUL}_t vs2, size_t vl);
15 //Move Vector -> Scalar
16 int{SEW}_t __riscv_vmv_x_s_i{SEW}{LMUL}_i{SEW}(vint{SEW}{LMUL}_t vs1);
17 //Move Scalar -> Vector
18 vint{SEW}{LMUL}_t __riscv_vmv_v_x_i{SEW}{LMUL}(int{SEW}_t rs1, size_t vl);
19 //Reduction
20 vint{SEW}m1_t __riscv_vredsum_vs_i{SEW}{LMUL}_i{SEW}m1(vint{SEW}{LMUL}_t vs2, vint{SEW}m1_t vs1,
21 size_t vl);
```